L14: Lesson Summary

This lesson focused on the overlap between Network Science and Machine Learning, with an emphasis on Deep Learning and Graph Neural Networks.

This is a relatively new area that has been mostly pursued in the research literature in the last five years or so.

The applications of this emerging area are numerous – mostly because many real-world problems can be modeled with graphs and because Deep Learning has enormous capabilities to learn complex features from data without the burden of manual ***"feature engineering".***

We should also mention however the main drawbacks of the Deep Learning approach:

First, the resulting models are over-parameterized *(thousands or even millions of parameters!)* – compare that with models such as Preferential Attachment that have only one parameter, or even Stochastic Block Graphs in which the number of parameters scales with the number of communities. Models with too many parameters may overfit the data and they can be computationally expensive in terms of training.

Second, Deep Learning models are often viewed as ***"black boxes",*** i.e., they are not transparent in terms of how the *(automatically identified)* features relate to the given task. For example, if a neural network classifies the node of an online social network as ***"bot"*** *(instead of human)*, we may not know why.

Third, Deep Learning models typically require lots of training data. This is not a problem as long as we are working with large graphs, or many graphs, and we have labeled data for the nodes and edges of those graphs. For smaller networks however, it may be more appropriate to rely on simpler models, such as those studied in Lesson-12.

The last part of this lesson also mentioned some other state-of-the-art network science topics (such as interdependent networks or temporal networks) that we do not have time unfortunately to cover in more detail. If you are interested to learn more about these topics, please refer to the provided references.